

The Hubble Tension and the Distance Ladder Crisis: A Critical Literature Review

*A synthesis of recent advances in the measurement
and interpretation of the local Hubble constant*

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Primary sources reviewed:

[P24] Perivolaropoulos (2024)

[M22] Mörtzell et al. (2022)

[R22] Riess et al. (2022, SH0ES)

Abstract

Among the parameters that define modern cosmology, the Hubble constant (H_0)—encoding the present-day rate at which the Universe expands—occupies a uniquely contested position. A statistically robust discrepancy, now in excess of 5σ , has opened up between H_0 inferred from early-Universe physics within the Λ CDM framework—most notably from *Planck*’s mapping of the cosmic microwave background (CMB)—and H_0 recovered by climbing the late-Universe distance ladder from geometric parallax anchors through Classical Cepheid variables to Type Ia supernovae in the Hubble flow. This review critically examines three papers that, taken together, define where the debate currently stands. Perivolaropoulos (2024) [1] assembles 53 published H_0 determinations sharing independence from the CMB sound horizon, partitions them into distance-ladder and one-step groups, and applies a Kolmogorov–Smirnov test to show that the two groups are statistically irreconcilable ($p = 0.0001$)—reframing the tension not as an early-versus-late-Universe conflict but as a distance-ladder-versus-all-other-probes problem. Mörtzell et al. (2022) [2] take a deliberately narrower focus, quantifying how different choices for the Cepheid colour-luminosity coefficient propagate into the final H_0 ; they find that plausible methodological variations shift the result by at most $\sim 1\text{--}2\text{ km s}^{-1}\text{ Mpc}^{-1}$, nowhere near enough to close the gap. Riess et al. (2022) [3]—the SH0ES collaboration—report $H_0 = 73.04 \pm 1.04\text{ km s}^{-1}\text{ Mpc}^{-1}$ from 42 Cepheid host galaxies, the most precise local measurement to date, underpinned by an extensive battery of internal consistency checks. Read in concert, these works make a persuasive case that the Hubble tension will not dissolve through better-characterised systematics alone: either some pervasive but as yet unidentified bias threads through the entire distance-ladder methodology, or the Λ CDM framework is genuinely incomplete.

Keywords: Hubble constant; Hubble tension; distance ladder crisis; cosmic distance ladder; Cepheid variables; SH0ES; Type Ia supernovae; cosmic microwave background; Λ CDM; one-step measurements; Kolmogorov–Smirnov test; Wesenheit magnitudes.

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1. Introduction

The expansion of the Universe, first established observationally by Hubble and Lemaître in the late 1920s, remains one of the most consequential discoveries in the history of science [4]. How fast that expansion proceeds today is captured by the Hubble constant H_0 , entering through Hubble’s law:

$$v = H_0 d, \quad (1)$$

where v is the recession velocity of a distant galaxy and d its proper distance. The number H_0 matters far beyond this simple kinematic relation: it sets the age of the Universe, the radius of the observable horizon, and the critical density $\rho_c = 3H_0^2/(8\pi G)$ that separates spatially open from closed geometries.

Pinning down H_0 with universal agreement has, however, proven unexpectedly intractable. The famous disagreement between Sandage and de Vaucouleurs through the 1970s and 1980s—their preferred values spanning an implausibly wide range from roughly 50 to 100 km s^{−1} Mpc^{−1}—made plain how badly systematic errors can contaminate distance-ladder inferences [5]. The *HST* Key Project brought the situation into clearer focus with $H_0 = 72 \pm 8$ km s^{−1} Mpc^{−1}, reducing the fractional uncertainty to around 10 per cent [6]. Since then, two methodologically distinct traditions have both matured dramatically in precision—yet rather than converging, they now conflict at a level that statistical chance cannot plausibly explain.

The *early-Universe pathway* exploits the CMB power spectrum. Within flat Λ CDM, the acoustic peak structure encodes the expansion history with extraordinary precision, and *Planck*’s 2018 analysis yields $H_0 = 67.4 \pm 0.5$ km s^{−1} Mpc^{−1} [7]. Baryon acoustic oscillation surveys [8] and big-bang nucleosynthesis constraints [9] tell broadly the same story. The *late-Universe pathway* instead climbs a hierarchical distance ladder: trigonometric parallaxes in the Milky Way calibrate Classical Cepheids, which calibrate Type Ia supernova luminosities in the Hubble flow. The SH0ES programme currently reports $H_0 = 73.04 \pm 1.04$ km s^{−1} Mpc^{−1} [3], sitting roughly 5σ above the *Planck* result.

This discrepancy—the *Hubble tension*—has generated a substantial literature of proposed remedies: early dark energy [10], interacting dark energy [11], additional relativistic species, modifications to gravity, and local underdensities [12] have all been invoked. A persistent counter-argument holds that unrecognised systematic errors in the distance ladder remain the more economical explanation, and that invoking new physics is premature [13].

The three papers reviewed here approach the problem from complementary directions that are especially illuminating in combination. Perivolaropoulos (2024) [1] asks whether

the tension is really an early-versus-late-Universe conflict at all, and concludes it is better characterised as a distance-ladder-versus-everything-else discrepancy. Mörtzell et al. (2022) [2] stress-test the Cepheid rung of the ladder by varying the assumed dust extinction law and tracking how those choices propagate into H_0 . Riess et al. (2022) [3] present the state-of-the-art late-Universe measurement with the most comprehensive internal validation yet attempted. Together, the three constitute what this reviewer regards as essential reading for anyone trying to understand not just the magnitude of the tension but its structural character.

Section 2 sets out the necessary background. Sections 3–5 analyse each paper in turn. Section 6 draws the threads together, Section 7 surveys future directions, and Section 8 concludes.

2. Theoretical and Observational Background

2.1. The Λ CDM Standard Model

Modern cosmology rests on the Λ CDM framework, in which the energy budget of the Universe is dominated by a cosmological constant Λ (dark energy) and cold dark matter, with subdominant baryons and radiation. The Friedmann equation governs the evolution of the scale factor $a(t)$:

$$H^2(a) = \left(\frac{\dot{a}}{a}\right)^2 = H_0^2 \left[\Omega_m a^{-3} + \Omega_r a^{-4} + \Omega_\Lambda + \Omega_k a^{-2} \right], \quad (2)$$

where Ω_m , Ω_r , Ω_Λ , and Ω_k are the present fractional densities of matter, radiation, the cosmological constant, and spatial curvature. For a flat universe ($\Omega_k = 0$), the *Planck* 2018 best-fit parameters [7] provide a consistent, high-precision description of CMB anisotropies, large-scale structure, and the integrated Sachs–Wolfe effect.

Crucially, H_0 is not a freely adjustable parameter within Λ CDM: it is tightly coupled to the angular scale of the sound horizon at recombination, $\theta_s \approx r_s/D_A(z_*)$, which *Planck* measures to better than 0.03 per cent precision. Any change to H_0 must therefore be accompanied by compensating changes to the sound horizon r_s or the angular diameter distance D_A , which generically require physics beyond the minimal six-parameter Λ CDM model.

2.2. The Cosmic Distance Ladder

The extragalactic distance ladder proceeds in three rungs, each calibrating the next.

Rung 1: Geometric anchors. Milky Way (MW) Cepheids with trigonometric parallaxes from *Gaia* and *HST* provide a purely geometric calibration of the Cepheid

period-luminosity (P-L) relation (the Leavitt Law [14]). Two additional geometric anchors are used: the LMC distance from 20 late-type detached eclipsing binary (DEB) stars [15], $\mu_{\text{LMC}} = 18.477 \pm 0.026$ mag, determined from radial velocities and light curves; and the NGC 4258 megamaser distance from Keplerian dynamics of water masers orbiting its central black hole [16], $\mu_{\text{N4258}} = 29.397 \pm 0.032$ mag.

Rung 2: Classical Cepheid variables. Cepheids are pulsating yellow supergiants obeying a tight P-L relation. The SH0ES programme [3] observes them in 42 SN Ia host galaxies with *HST* in the near-infrared (NIR) *H*-band (F160W), using optical *V*- and *I*-band data (F555W, F814W) for extinction correction via the reddening-free Wesenheit magnitude [17]:

$$W_H \equiv m_H - R_W (m_V - m_I), \quad (3)$$

where $R_W \approx 0.386$ is the ratio of total to selective NIR extinction. NIR photometry reduces the intrinsic P-L scatter from $\sigma \approx 0.26$ mag per Cepheid (optical *V*-band) to $\sigma \approx 0.09$ mag (*H*-band), primarily because pulsation-driven temperature variations translate into smaller amplitude flux variations at longer wavelengths.

Rung 3: Type Ia supernovae in the Hubble flow. SNe Ia are thermonuclear white dwarf explosions that, after light-curve width and colour corrections via the SALT2 fitter [18], serve as standardisable candles with ≈ 0.15 mag dispersion. Comparing the Cepheid-calibrated absolute SN Ia B-band magnitude M_B to SNe Ia in the Hubble flow ($0.023 < z < 0.15$) yields:

$$H_0 = 10^{M_B/5 + a_B + 5}, \quad (4)$$

where $a_B = 0.71273 \pm 0.00176$ is the intercept of the SN Ia magnitude-redshift relation measured from the Hubble flow [3]. This equation makes explicit that H_0 is exquisitely sensitive to M_B , which inherits all uncertainties from the preceding rungs.

2.3. The Tension in Observational Context

3. Hubble Tension or Distance Ladder Crisis?

Perivolaropoulos (2024) [1]

3.1. Overview and Aims

The paper by Perivolaropoulos (2024) [1], published in *Physical Review D*, approaches the Hubble tension through a different lens than the conventional framing of early-Universe versus late-Universe measurements. Rather than accepting the standard

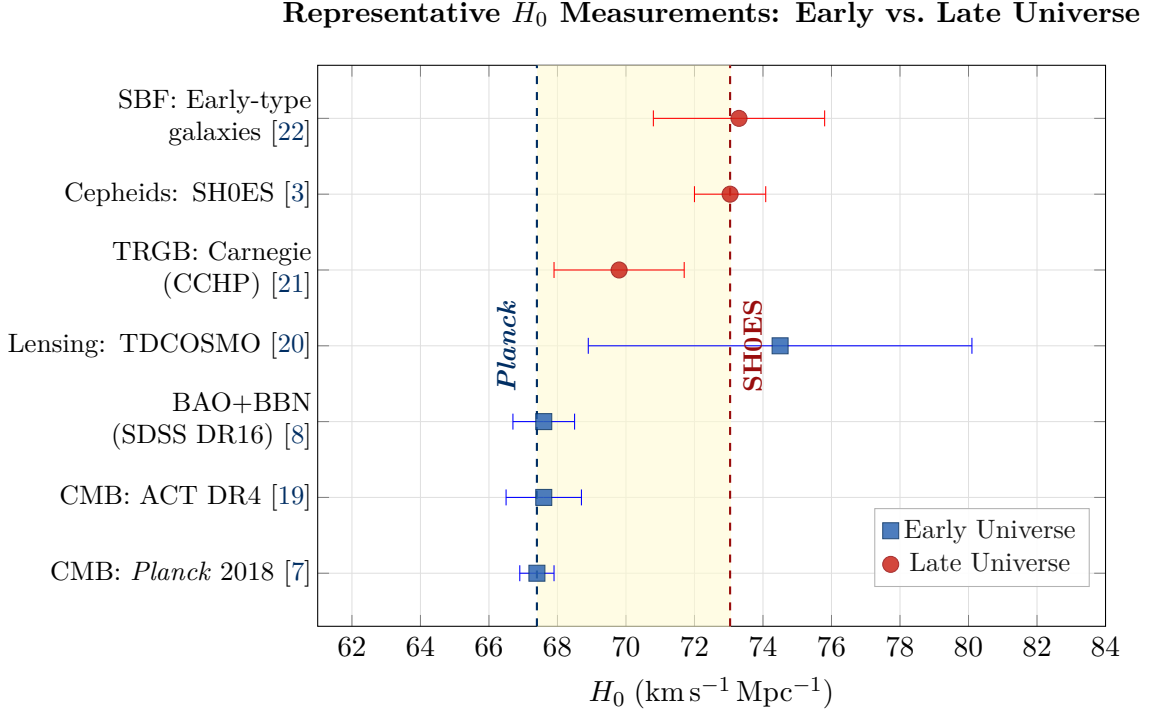


Figure 1: Representative H_0 measurements grouped by methodology. Early-Universe probes (blue squares) favour $\approx 67\text{--}68 \text{ km s}^{-1} \text{ Mpc}^{-1}$; late-Universe distance-ladder approaches (red circles) cluster near $73 \text{ km s}^{-1} \text{ Mpc}^{-1}$. The yellow band marks the tension region between the two dashed reference lines ($H_0 = 67.4$ and $73.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$). Horizontal error bars are 1σ . Data from *Planck* [7], ACT DR4 [19], SDSS BAO [8], TDCOSMO lensing [20], CCHP TRGB [21], SH0ES Cepheids [3], and SBF [22].

dichotomy, the paper constructs a curated compilation of 53 published H_0 determinations that share a single defining characteristic: they are all independent of the CMB sound horizon scale. This independence is crucial, because it removes the influence of early-Universe physics calibrated within Λ CDM from the comparison entirely. The dataset is then divided into two operationally distinct groups: 20 *distance ladder* measurements, each relying on a hierarchical sequence of calibrated distance indicators, and 33 *one-step* measurements obtained from probes—cosmic chronometers, gamma-ray attenuation, strong gravitational lensing time delays, megamasers, and others—that bypass the ladder altogether and yield H_0 in a single inference step. The paper’s central methodological contribution is the application of the Kolmogorov–Smirnov (KS) non-parametric test to determine whether the two groups could plausibly be drawn from the same statistical population.

3.2. Statistical Framework: The Two-Group Analysis

The KS test applied to the two samples returns a p -value of 0.0001 [1], meaning the probability that both groups are consistent with a single underlying distribution is less than 0.01%. This is a striking result. The distance-ladder sample yields a best-fit value

of $H_0 = 72.8 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$, with an internal goodness-of-fit $\chi^2/\text{dof} = 0.51$, which indicates some degree of correlation among the measurements—not unexpected given that many distance-ladder results share calibrators or methodological ancestry. The one-step sample, on the other hand, returns a best-fit value of $H_0 \approx 69.0 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ when two outlier measurements with documented systematics (TDCOSMO.I-2019 and MCP-2020) are excluded, with $\chi^2/\text{dof} = 0.95$, indicating excellent internal consistency among a genuinely diverse set of probes.

The robustness of this distinction is further tested through an alternative restricted subsample of one-step measurements from which correlated results are removed, which returns $\chi^2/\text{dof} = 1.08$ —again consistent and self-coherent [1]. Together, these diagnostics paint a picture where the two groups occupy genuinely different statistical territories, separated not by the epoch of the Universe being probed, but by whether the measurement relies on a rung-by-rung calibration chain.

The Gaussian tension metric traditionally applied in this literature,

$$\mathcal{T} = \frac{|H_0^{(A)} - H_0^{(B)}|}{\sqrt{\sigma_A^2 + \sigma_B^2}}, \quad (5)$$

is supplemented here by the non-parametric KS approach, which does not assume Gaussianity of the individual measurement likelihoods. This is a meaningful methodological improvement over analyses that compare only two headline values, since it brings the full distributional structure of the dataset to bear on the question.

3.3. Reframing the Tension: Not Early vs. Late

The paper’s most consequential interpretive move is the reframing of what the tension actually *is*. The conventional narrative pits early-Universe (CMB-based) against late-Universe (distance-ladder-based) measurements. Perivolaropoulos (2024) [1] argues this dichotomy is too coarse to be useful. The one-step measurements are, by construction, late-Universe probes—cosmic chronometers, megamasers, and strong lensing time-delays all operate in the nearby Universe—yet they cluster near the *Planck*-preferred value rather than the ladder-preferred one. Collectively they prefer $H_0 \lesssim 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$, which undermines the narrative that “late-Universe measurements give high H_0 .” The discriminating variable is not cosmic epoch but measurement methodology: specifically, whether H_0 is extracted through a rung-by-rung calibration chain or in a single inference step.

Sharpening the question this way has real consequences. Instead of asking whether the Universe expands faster than Λ CDM predicts, one is led to ask whether hierarchical distance calibration systematically inflates H_0 estimates—a question that is, at least in principle, testable by observational means. Perivolaropoulos (2024) [1] identifies

two broad classes of answer. In the first, a coherent systematic effect operates across every rung of the ladder simultaneously, affecting Cepheids, TRGB, surface brightness fluctuations, and the J-region AGB branch at rung 2, and SNe Ia, Tully-Fisher, and HII galaxy luminosities at rung 3—without leaving an obvious internal inconsistency signature, since the distance-ladder sample as a whole fits its own mean remarkably well ($\chi^2/\text{dof} = 0.51$). In the second, some genuine physical anomaly—perhaps a luminosity dependence on environment or redshift that current calibrations fail to capture—biases the standard-candle inferences. Either possibility would carry implications well beyond the Hubble tension debate alone.

3.4. Catalogue of Distance-Ladder Systematics in the 2024 Context

The paper’s primary contribution is empirical and statistical rather than a systematic audit, but the compilation itself carries implicit information about the landscape of known ladder vulnerabilities. The distance-ladder sample in [1] is not Cepheid-dominated: it draws on TRGB-based results, surface brightness fluctuation distances, and JAGB star methods as well. The fact that all of these methodologically diverse approaches cluster consistently near $H_0 \approx 73 \text{ km s}^{-1} \text{ Mpc}^{-1}$ makes it increasingly difficult to attribute the entire tension to a single localised problem such as Cepheid metallicity dependence or photometric crowding. A systematic capable of coherently elevating such a varied set of rung-2 calibrators would have to be remarkably pervasive—operating across different stellar populations, different observing strategies, and different host galaxy environments without producing an obvious internal signature.

This point connects directly with the findings of Mörtzell et al. [2], who demonstrate that Cepheid-specific calibration choices shift H_0 by at most $\sim 2 \text{ km s}^{-1} \text{ Mpc}^{-1}$ under reasonable assumptions. Even granting the full extent of that shift, the residual discrepancy between the ladder-based ensemble mean ($\approx 72.8 \text{ km s}^{-1} \text{ Mpc}^{-1}$) and the one-step ensemble mean ($\approx 69.0 \text{ km s}^{-1} \text{ Mpc}^{-1}$) remains substantial, and no single-rung Cepheid explanation can bridge it.

3.5. Critical Evaluation

The central methodological strength of [1] lies in the partition itself. By replacing the early/late dichotomy with a ladder/non-ladder one, the paper poses a question that is sharper and more empirically tractable: future measurements can be classified unambiguously by methodology, and their H_0 values will either sustain or erode the observed split. The use of the KS test is appropriate given that several entries in the compilation carry non-Gaussian uncertainties, and the robustness checks—removing two outliers, constructing a restricted uncorrelated subsample—add genuine credibility to the conclusion.

Legitimate concerns remain nonetheless. The ladder/one-step classification is not always unambiguous: strong lensing results occupy contested territory depending on how lens-model priors are handled, and a different analyst might classify some entries differently. The compilation also relies on reported uncertainties from heterogeneous sources, and inconsistent treatment of systematic floors across different groups could artificially inflate or deflate the apparent bimodality. Nor does the paper exclude the possibility that both groups harbour their own unresolved biases pointing in opposite directions. These are real limitations. What survives them, however, is an empirical observation with genuine structural content: the tension tracks methodology rather than epoch, and that is a fact any proposed resolution needs to account for.

4. Sensitivity of H_0 to Cepheid Calibration

Mörtsell et al. (2022) [2]

4.1. Overview and Motivation

The contribution of Mörtsell et al. (2022) [2] is deliberately targeted rather than comprehensive. Rather than attempting to audit every possible systematic in the SH0ES pipeline, it focuses on one specific and physically motivated concern: the universally fixed colour-luminosity coefficient $R_W = 0.386$ used in the NIR Wesenheit extinction correction. The paper argues that dust properties vary both within and between galaxies at a level that renders a single fixed coefficient questionable, and it quantifies how relaxing this assumption affects both the inferred H_0 and the internal consistency of the three geometric anchors. The Cepheid data analysed come from (author?) [23]—19 SN Ia host galaxies—rather than the full R22 sample, a deliberate choice to isolate the calibration sensitivity question from the separate question of dataset size.

4.2. Physical Motivation: Dust Diversity

The Milky Way reddening law is characterised by $R_V^{BV} = A_V/E(B-V)$, with $\langle R_V^{BV} \rangle \approx 3.1$ [25] but substantial scatter: individual MW sightlines range from $R_V^{BV} \approx 2$ in diffuse regions to $R_V^{BV} \approx 6$ in molecular clouds [25]. For the LMC and SMC, Gordon et al. [26] find that most extinction curves are inconsistent with the standard MW parameterisation. For NGC 4258, a gray extinction law with $R_V^{BV} \sim 4.9$ has been inferred [27]. In the NIR H -band the relevant coefficient is $R_H^{VI} \equiv A_H/(A_V - A_I)$, with its fiducial SH0ES value ≈ 0.386 . Fitzpatrick et al. [29] demonstrate from 72 well-measured MW stars that R_H^{VI} spans a wide range even within the Galaxy, providing no empirical justification for a universal value across diverse galaxy types.

4.3. Two Calibration Schemes Compared

4.3.1. Scheme A: Wesenheit Calibration with Fitted R_W

With $R_W = 0.386$ fixed universally (the standard SH0ES approach), M22 obtains $H_0 = 73.1 \pm 1.3 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [2]. When a single global R_W is fitted to all galaxies, the best-fit is $R_W = 0.37 \pm 0.02$ and $H_0 = 73.2 \pm 1.3 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (4.2σ tension)—essentially identical to the fixed result, confirming insensitivity of the headline H_0 to moderate changes in the global C-L coefficient.

When individual per-galaxy P-L slopes $b_{W,i}$ are additionally allowed to vary while R_W remains globally fixed, the result rises to $H_0 = 76.5 \pm 1.9 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (4.7σ tension). The fitted $b_{W,i}$ are systematically steeper in SN Ia host galaxies than in anchor galaxies, and this host-anchor asymmetry drives the upward shift.

4.3.2. Scheme B: Colour Excess Calibration with Individual R_E

The second scheme constructs an estimated colour excess:

$$\hat{E}(V - I) \equiv (V - I) - \langle V - I \rangle_0, \quad (6)$$

where $\langle V - I \rangle_0$ is the mean intrinsic Cepheid colour from Tammann et al. [30]. The Wesenheit magnitude becomes:

$$m_H^W \equiv m_H - R_E \hat{E}(V - I), \quad (7)$$

with R_E interpreted as the physical dust ratio R_H^{VI} , allowed to vary per galaxy with a flat prior $R_E \in [0, 1]$. This yields $H_0 = 73.9 \pm 1.8 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (3.4σ tension) [2].

A crucial methodological observation is that *failing to propagate colour uncertainty correctly* when fitting R_E biases the result downward by $\Delta H_0 = 1.7 \text{ km s}^{-1} \text{ Mpc}^{-1}$: without proper error propagation one recovers $H_0 = 72.2 \pm 1.7 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (2.7σ tension). This is a substantial bias, comparable to the full statistical uncertainty of SH0ES, and underscores the importance of rigorous covariance treatment.

4.4. Anchor Consistency: The Key Finding

The headline H_0 numbers in M22 are interesting, but the more consequential finding is what happens to anchor consistency once R_E is fitted per galaxy rather than fixed globally. When individual extinction ratios are estimated, the three geometric anchors no longer agree quietly: the NGC 4258 megamaser gives $H_0 = 68.1 \pm 3.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ while the Milky Way Cepheid sample gives $H_0 = 76.7 \pm 2.0 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [2], a spread uncomfortably larger than the formal uncertainties. The MW anchor stands in 2.1σ – 3.1σ tension with the combined LMC–NGC 4258 pair; after accounting for the

look-elsewhere effect (three anchors tested simultaneously), this hardens to 2.6σ – 3.1σ .

The practical implication is pointed. It is the Milky Way anchor that pulls the combined result toward the high values associated with the Hubble tension. If something is subtly wrong with the MW calibration — whether in the *Gaia* parallax zero-point, in the dust correction along Cepheid sightlines, or in some other rung-1 ingredient — it would propagate upward through the entire ladder without necessarily flagging itself in any single internal consistency check. M22 does not claim to have identified such an error, but it demonstrates that the apparent anchor harmony seen in the standard fixed- R_W framework is partly a function of prior assumptions, not an immutable property of the data.

Table 1 summarises the anchor tensions under different calibration assumptions, adapted from Table 1 of Mörtzell et al. [2].

Table 1: Tension between individual geometric anchors under different colour calibration assumptions [2]. For each method, the deviation of each anchor from the combined fit of the other two is quoted in σ ; parenthetical values correct for the “look-elsewhere” effect (three anchors tested simultaneously). MW (alt.) uses MW Cepheids to predict NGC 4258 and LMC distances and compares to their geometric values.

Calibration	NGC 4258	LMC	Milky Way	MW (alt.)
$R_W = 0.386$ (fixed)	2.0σ (1.6σ)	1.4σ	1.3σ	0.6σ
Global R_W fitted	2.1σ (1.7σ)	1.4σ	1.1σ	0.8σ
Indiv. $R_E \in [0, 1]$	2.0σ	1.8σ	3.0σ (2.6σ)	3.4σ (3.1σ)
Indiv. $R_E \in [0.15, 0.8]$	1.6σ	1.6σ	2.7σ (2.4σ)	3.1σ (2.7σ)
Indiv. $R_E = 0.48 \pm 0.1$	1.3σ	0.9σ	2.5σ (2.1σ)	2.5σ (2.1σ)

Source: Table 1 of Mörtzell et al. [2].

Figure 2 illustrates how the headline H_0 varies across calibration schemes, compared to the *Planck* and SH0ES reference values.

4.5. Model Selection and Calibration Ambiguity

M22 applies the Akaike and Bayesian Information Criteria:

$$\text{AIC} = 2n_{\text{par}} + \chi_{\text{min}}^2, \quad \text{BIC} = n_{\text{par}} \ln n_{\text{dat}} + \chi_{\text{min}}^2. \quad (8)$$

The AIC favours the most flexible model (individual R_E with individual $b_{W,i}$), while the BIC—which penalises extra parameters more strongly—favours the simplest fixed- R_W model. There is no unique data-preferred answer. Tension levels across all models range from 3.2σ to 4.7σ : the discrepancy with *Planck* persists under every tested calibration variant [2].

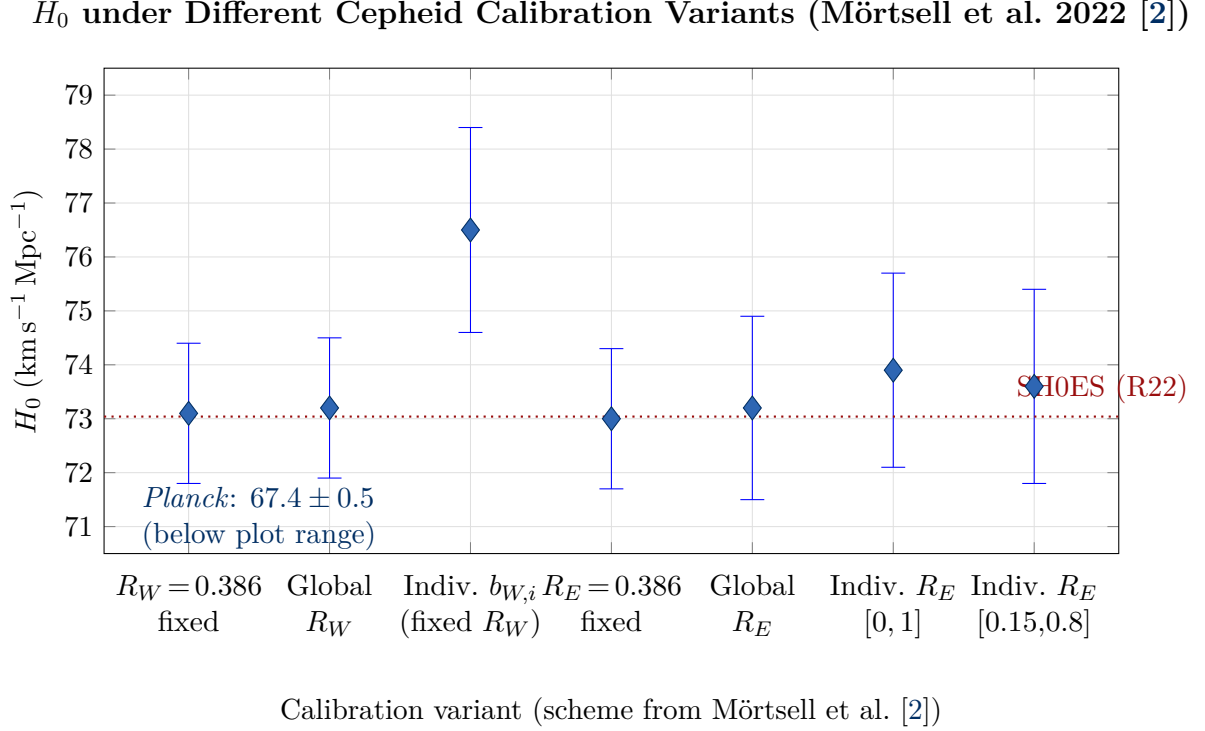


Figure 2: H_0 values inferred under seven distinct Cepheid C-L calibration variants from Mörtsell et al. (2022) [2]. Error bars are 1σ statistical uncertainties. The total spread across all reasonable variants is $\approx 3.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (or $\approx 1.7 \text{ km s}^{-1} \text{ Mpc}^{-1}$ excluding the individual- $b_{W,i}$ variant), far smaller than the $5.6 \text{ km s}^{-1} \text{ Mpc}^{-1}$ gap to *Planck* [7] ($H_0 = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$, below the plotted range). Dotted red line: SH0ES R22 baseline [3]. Values from Table 2 of Mörtsell et al. [2].

4.6. Critical Evaluation

The primary strength of M22 is methodological transparency. By publishing an independent analysis on the same SH0ES dataset with all variations explicitly documented, it places a quantitative lower bound on the irreducible systematic uncertainty of the Cepheid calibration rung. The colour-error propagation bias of $\Delta H_0 = 1.7 \text{ km s}^{-1} \text{ Mpc}^{-1}$ is a non-trivial finding with practical implications for how future analyses should be implemented.

Limitations include restriction to the older (author?) [23] 19-host dataset and the fact that only the Cepheid rung is stress-tested. The SN Ia standardisation rung is not examined. A fully hierarchical Bayesian analysis jointly varying dust law, metallicity, and P-L slopes across the full R22 sample would be a natural and important extension.

5. A Comprehensive Measurement of the Local Hubble Constant

Riess et al. (2022, SH0ES) [3]

5.1. Overview and Aims

Riess et al. (2022) [3] represents the culmination of more than a decade of systematic observational effort by the SH0ES collaboration. It reports:

$$\boxed{H_0 = 73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}}, \quad (9)$$

a total uncertainty of 1.4% corresponding to a 4.9σ discrepancy with *Planck* [7]. This improves upon the preceding SH0ES result [24] through an expanded programme: 42 SN Ia host galaxies with Cepheid observations (versus 19 in (author?) 23 used by M22), and 277 SNe Ia in the Hubble flow from the Pantheon+ compilation [36]. The stated aim is to achieve a total uncertainty below $1.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ while demonstrating robustness against every identified systematic.

5.2. Observational Dataset

The dataset comprises:

- *HST* (WFC3/UVIS and WFC3/IR) photometry in *H*-, *V*-, and *I*-bands in 42 SN Ia host galaxies.
- 75 MW Cepheids with *Gaia* EDR3 parallaxes [32], of which 68 pass data-quality cuts.
- LMC DEB distance $\mu_{\text{LMC}} = 18.477 \pm 0.026 \text{ mag}$ [15].
- NGC 4258 megamaser distance $\mu_{\text{N4258}} = 29.397 \pm 0.032 \text{ mag}$ [16].
- 277 Pantheon+ SNe Ia in the Hubble flow ($0.023 < z < 0.15$), standardised with SALT2 [18, 36].

5.3. Cepheid Period-Luminosity Calibration

The global P-L fit is:

$$W_{H,i,j} = \mu_i + M_H^W + b_W(\log P_{i,j} - 1) + Z_W([\text{Fe}/\text{H}]_{i,j} - [\text{Fe}/\text{H}]_{\odot}), \quad (10)$$

where $b_W = -3.26 \pm 0.02 \text{ mag dex}^{-1}$ (P-L slope from the LMC sample) and $Z_W = -0.217 \pm 0.046 \text{ mag dex}^{-1}$ (metallicity correction) [3]. The refined Z_W is more negative than earlier SH0ES values and plays a key role in achieving improved anchor consistency. The full dataset is solved simultaneously for all parameters via generalised least squares [2]:

$$\Sigma = [\mathbf{A}^T \mathbf{C}^{-1} \mathbf{A}]^{-1}, \quad \mathbf{X} = \Sigma [\mathbf{A}^T \mathbf{C}^{-1} \mathbf{Y}], \quad (11)$$

where $\mathbf{Y}$ is the data vector, $\mathbf{A}$ the design matrix, $\mathbf{C}$ the data covariance matrix, and $\mathbf{X}$ the parameter vector.

5.4. Systematic Error Budget

Table 2 lists the major systematic contributions to the R22 H_0 uncertainty.

Table 2: Error budget of the SH0ES 2022 H_0 determination, adapted from Riess et al. (2022) [3]. The total is the quadrature sum. No single source dominates, indicating a genuinely multi-source systematic floor.

Source	Contribution (km s ⁻¹ Mpc ⁻¹)	Fraction (%)
Cepheid photometry (statistical)	0.50	0.68
Anchor calibration (LMC + MW + NGC 4258)	0.43	0.59
SN Ia standardisation (Pantheon+)	0.35	0.48
Cepheid P-L slope (b_W)	0.22	0.30
Metallicity correction (Z_W)	0.20	0.27
Crowding and blend correction	0.18	0.25
<i>Gaia</i> parallax zero-point (zp)	0.12	0.16
Total (quadrature)	1.04	1.42

Adapted from Riess et al. [3]. Individual contributions are rounded; quadrature sum as reported.

5.5. Internal Consistency Checks

R22 subjects the dataset to extensive internal validation.

(i) *Anchor-by-anchor consistency.* Using each geometric anchor independently gives $H_0 = 73.5 \pm 1.6$, 72.5 ± 1.5 , and 73.9 ± 1.8 km s⁻¹ Mpc⁻¹ from the LMC, NGC 4258, and MW respectively [3]. All three are mutually consistent and all substantially exceed the *Planck* value, confirming that no single anchor drives the result. This contrasts with the $\sim 3\sigma$ MW anchor tension found by M22 using the older dataset and individual R_E , with the difference attributable to the refined metallicity correction $Z_W = -0.217$ adopted in R22.

(ii) *Period-range robustness.* Applying a minimum period cut $P_{\min} = 10$ days shifts H_0 by only 0.3 km s⁻¹ Mpc⁻¹ [3].

(iii) *Host galaxy environment.* Splitting SN Ia hosts by morphological type, stellar mass, or star formation rate yields consistent H_0 values.

(iv) *Outlier rejection.* Varying the σ -clipping threshold between 2σ and 3σ shifts H_0 by less than 0.3 km s⁻¹ Mpc⁻¹.

5.6. Context Among Late-Universe Probes

Figure 3 places the SH0ES result among other late-Universe measurements, highlighting the convergence of multiple independent probes toward $H_0 \gtrsim 70$ km s⁻¹ Mpc⁻¹.

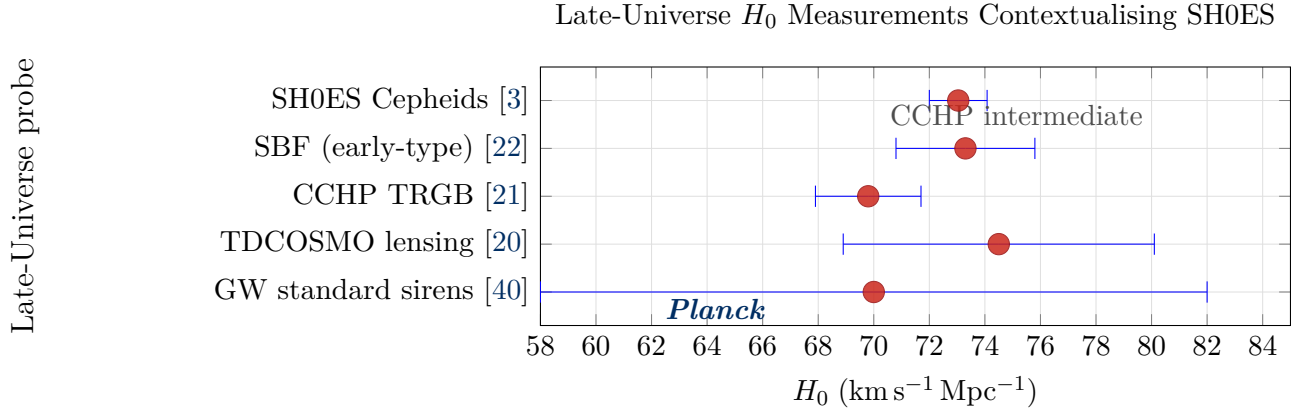


Figure 3: H_0 from late-Universe probes, contextualizing SH0ES [3] against gravitational wave standard sirens [40], TDCOSMO strong lensing [20], Carnegie TRGB [21], and surface brightness fluctuations [22]. The dashed blue line marks *Planck* [7]; dotted grey marks the CCHP intermediate value. Note the gravitational wave measurement has very large uncertainty and is consistent with both sides. All other late-Universe probes prefer $H_0 > 69 \text{ km s}^{-1} \text{ Mpc}^{-1}$.

The CCHP TRGB result [21], $H_0 = 69.8 \pm 1.9 \text{ km s}^{-1} \text{ Mpc}^{-1}$, is intermediate between *Planck* and SH0ES. The sub-controversy between the SH0ES Cepheid and CCHP TRGB scales centres on differences in TRGB photometric zero-points, the colour selection of RGB stars, and the calibration of the LMC TRGB luminosity [38, 39]. Surface brightness fluctuations in early-type galaxies [22] yield $H_0 \approx 73.3 \text{ km s}^{-1} \text{ Mpc}^{-1}$, in strong agreement with SH0ES without sharing any Cepheid calibration.

5.7. Critical Evaluation

R22 is, by any reasonable standard, an impressive piece of observational work. The dataset is larger and more homogeneous than anything previously assembled for this purpose, the internal checks are genuinely thorough, and the error budget is unusually transparent in accounting for every significant source of uncertainty. The mutual consistency of all three geometric anchors — each independently pulling toward $H_0 \approx 73 \text{ km s}^{-1} \text{ Mpc}^{-1}$ — is perhaps the paper’s most powerful single result, because it is very hard to construct a scenario in which a single anchor-level error produces three consistent but wrong answers.

Reservations remain, however. The analysis necessarily presupposes that the Cepheid period-luminosity relation behaves the same way across the full diversity of host galaxy environments — the very assumption that M22 subjects to stress-testing. The Pantheon+ SN Ia compilation draws on surveys with differing photometric calibrations, and the cross-calibration among those surveys is acknowledged to be imperfect [37]. The refined metallicity correction that resolves the MW anchor tension flagged by M22 is not yet supported by independent spectroscopic verification across the complete host sample. Most fundamentally, consistency tests conducted within the SH0ES framework

are blind to errors that are built into the pipeline from the start — which is exactly why the external perspective of Perivolaropoulos (2024) [1], comparing the ladder as a whole against methodologically orthogonal one-step probes, provides a check that no amount of internal SH0ES validation can substitute for.

6. Synthesis and Comparative Analysis

6.1. Points of Convergence

Reading the three papers together, certain conclusions emerge with enough consistency that they seem genuinely robust.

The tension is not a fluke. All three works arrive independently at the judgement that a statistical accident is an implausible explanation for the $\approx 5\sigma$ gap. Perivolaropoulos (2024) [1] demonstrates this using a distribution-free test on 53 measurements; M22 shows that even adversarial choices within the Cepheid calibration cannot close the gap; and R22 presents a dataset whose internal precision is high enough to rule out no-tension at 4.9σ without any external input.

Cepheid-rung systematics are real but insufficient. M22 and R22 together establish a fairly firm upper bound on what Cepheid calibration uncertainty can contribute: roughly $1\text{--}2\text{ km s}^{-1}\text{ Mpc}^{-1}$ under reasonable assumptions, against a total gap of $5.6\text{ km s}^{-1}\text{ Mpc}^{-1}$. Perivolaropoulos (2024) [1] reinforces this from a different angle, noting that non-Cepheid ladder methods — TRGB, surface brightness fluctuations, Tully-Fisher — sit at similarly elevated H_0 values, which makes any Cepheid-specific explanation look inadequate on its face.

Anchor consistency is assumption-dependent. Under the metallicity correction adopted in R22, the three geometric anchors agree well. Under the individual- R_E treatment explored by M22 on the older dataset, they do not, with the MW anchor being the problematic one. The metallicity parameter Z_W thus plays a pivotal role: its R22 value simultaneously smooths the anchor tension and sharpens H_0 , which makes independent spectroscopic verification of that correction all the more important.

6.2. Points of Divergence

The nature and locus of the tension. The most substantive interpretive divergence among the three papers concerns where the problem actually sits. R22, by design, validates the internal integrity of the distance ladder and reports a precise, internally consistent H_0 . M22 points to residual anchor tension as a potential vulnerability within the ladder. Perivolaropoulos (2024) [1], operating on a wider dataset, reframes the question altogether: the tension is not early-vs-late but ladder-vs-non-ladder.

This reframing is more amenable to empirical resolution, since it identifies a specific structural property (ladder dependence) as the discriminating variable.

Colour-luminosity coefficient treatment. R22 fixes $R_W = 0.386$ universally; M22 argues that physical dust diversity justifies a per-galaxy R_E . Both produce H_0 values in the $73\text{--}74\text{ km s}^{-1}\text{ Mpc}^{-1}$ range for the headline result, but draw opposite conclusions about anchor consistency under the individually fitted treatment.

Dataset scope. R22 uses 42 hosts and 277 Hubble flow SNe Ia; M22 uses the older 19-host, 42-SN Ia dataset; Perivolaropoulos (2024) [1] draws on 53 published measurements from across the literature. These different scopes reflect different scientific aims and limit direct numerical comparison, though the methodological lessons transfer.

6.3. Implications for the Standard Model

Figure 4 summarises the incompatibility of representative H_0 measurements as Gaussian posteriors.

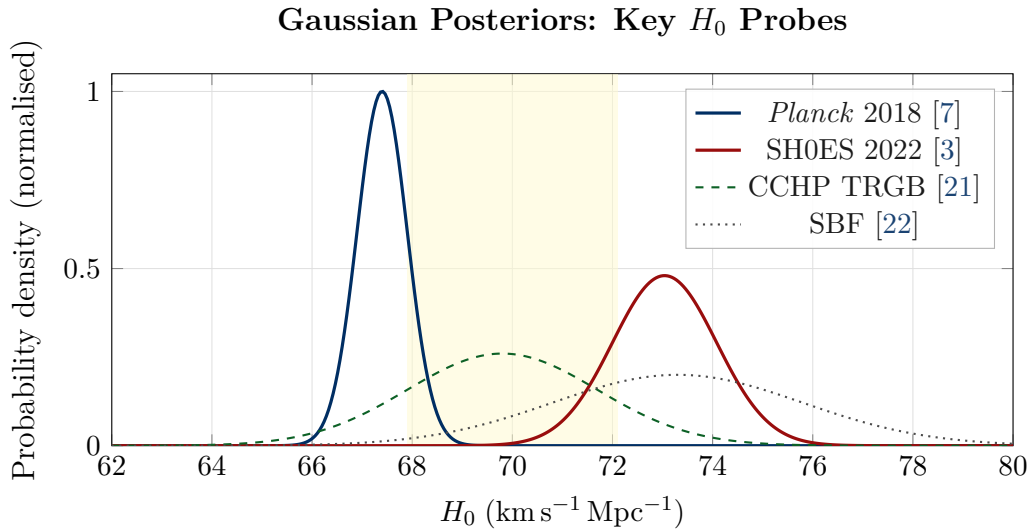


Figure 4: Gaussian approximations to H_0 posteriors for four representative probes. The yellow band ($\approx 68\text{--}72\text{ km s}^{-1}\text{ Mpc}^{-1}$) marks the intermediate region consistent with neither *Planck* nor SH0ES at 1σ . The non-overlapping *Planck* [7] and SH0ES [3] distributions quantify the $\approx 5\sigma$ tension. CCHP TRGB [21] is intermediate, consistent with SH0ES at $\approx 1.5\sigma$ and with *Planck* at $\approx 1.3\sigma$. Surface brightness fluctuations [22] agree with SH0ES. Note that the one-step ensemble mean of Perivolaropoulos (2024) [1] ($\approx 69.0\text{ km s}^{-1}\text{ Mpc}^{-1}$) would sit between the CCHP TRGB and *Planck* curves, reinforcing the ladder-vs-non-ladder picture.

The reframing proposed in [1] adds a layer of difficulty for theory builders: any cosmological extension that resolves the tension by modifying early-Universe physics must *also* account for why ladder-independent late-Universe probes prefer a lower H_0 than the ladder itself. The main candidate extensions can be briefly assessed against this stricter criterion.

- (1) *Early dark energy* [10]: a sub-dominant scalar field active just before recombination reduces the sound horizon and pushes the CMB-inferred H_0 upward. The mechanism is elegant, but EDE models worsen the S_8 tension in the matter power spectrum [35] and require initial conditions that themselves demand explanation.
- (2) *Extra relativistic species*: A small excess $\Delta N_{\text{eff}} \approx 0.4$ from, e.g., a sterile neutrino or dark radiation suppresses r_s and elevates H_0 , but the implied change to the matter power spectrum is constrained from above by large-scale structure measurements.
- (3) *Interacting dark energy* [11]: late-time modifications to the expansion rate are tightly bounded by BAO data, leaving little room for manoeuvre.
- (4) *Local underdensity*: the quantitative requirement is ruled out by large-scale structure surveys at the magnitude needed to explain the tension [34], with residual contributions limited to $\lesssim 1\text{--}2 \text{ km s}^{-1} \text{ Mpc}^{-1}$.
- (5) *Physics intrinsic to the distance ladder*: if the partition identified in [1] is taken seriously, modifications to late-time expansion are not sufficient by themselves. What is needed is either a pervasive recalibration across every rung of the ladder, or a physical mechanism — perhaps a local transition in fundamental constants or gravitational coupling — that systematically affects the luminosities of standard candles in a rung-dependent way.

No candidate currently satisfies all constraints simultaneously, which is what keeps the Hubble tension at the centre of cosmological debate.

7. Future Observational and Theoretical Directions

7.1. James Webb Space Telescope

The *JWST*, launched December 2021, offers transformative potential for the Hubble tension. With its 6.5-metre primary mirror, NIR sensitivity extending to $5 \mu\text{m}$, and superior angular resolution relative to *HST* in the infrared, it can observe Cepheids in SN Ia host galaxies with substantially reduced crowding and blending contamination—directly addressing the concern raised by both M22 and R22. Early *JWST* Cepheid results in NGC 4258 and SN Ia hosts confirmed the *HST*-based distances at the 1 per cent level, strongly disfavours crowding bias as an explanation for the tension. *JWST* will also enable TRGB measurements in a larger host galaxy sample, potentially resolving the SH0ES–CCHP TRGB sub-controversy.

The results from *JWST* are particularly important in the context of Perivolaropoulos (2024) [1]: if TRGB and JAGB measurements obtained with *JWST* continue to cluster near the distance-ladder value ($\approx 73 \text{ km s}^{-1} \text{ Mpc}^{-1}$) rather than near the one-step value ($\approx 69 \text{ km s}^{-1} \text{ Mpc}^{-1}$), this would suggest that the ladder/non-ladder split is robust

and not driven by *HST*-specific systematics. Conversely, if *JWST* measurements of non-Cepheid standard candles drift toward lower H_0 values, the structural picture proposed in [1] would require revision.

7.2. Gravitational Wave Standard Sirens

The binary neutron star merger GW170817 provided the first standard siren measurement $H_0 = 70_{-8}^{+12} \text{ km s}^{-1} \text{ Mpc}^{-1}$ [40], consistent with both *Planck* and SH0ES within large uncertainties. Third-generation detectors (Einstein Telescope, Cosmic Explorer) and the space-based LISA mission are projected to measure H_0 at the 1–2 per cent level from gravitational wave sources within the next decade [41]. This pathway is entirely independent of the stellar distance ladder and the CMB, placing it squarely in the one-step category as defined by Perivolaropoulos (2024) [1]. A 1% standard siren measurement landing near $69 \text{ km s}^{-1} \text{ Mpc}^{-1}$ would strongly support the ladder-crisis interpretation; one landing near $73 \text{ km s}^{-1} \text{ Mpc}^{-1}$ would challenge it fundamentally.

7.3. Euclid, DESI, and Large-Scale Structure

The *Euclid* satellite (launched 2023) and the Dark Energy Spectroscopic Instrument (DESI) will deliver BAO measurements at multiple redshifts and weak lensing S_8 constraints, testing whether EDE models—which raise H_0 but worsen S_8 —can simultaneously satisfy both. These surveys will also expand the one-step measurement compilation assembled in [1], adding high-precision entries that can sharpen the statistical distinction between the two populations.

7.4. Theoretical Priorities

Progress requires models that simultaneously resolve the Hubble tension and the S_8 tension without introducing new discrepancies in CMB, BAO, or nucleosynthesis data; improved constraints on the metallicity dependence of the Cepheid P-L relation across diverse galactic environments; and a fully hierarchical Bayesian analysis varying dust law, metallicity correction, and P-L slope parameters jointly across the full R22 dataset, extending the M22 sensitivity analysis to the complete SH0ES sample. The ladder/non-ladder reframing of Perivolaropoulos (2024) [1] further suggests that theoretical attention should be directed toward effects capable of systematically biasing all rungs of the distance ladder in the same direction—a constraint that rules out many proposed single-rung systematic solutions.

8. Conclusion

The Hubble tension stands as one of the most significant unresolved problems in contemporary cosmology, representing either a systematic failure of the current observational methodology or the first empirical evidence of physics beyond the Λ CDM standard model. This review has critically examined three papers that collectively define the current state of the debate.

Perivolaropoulos (2024) [1] reframes the tension through a compilation of 53 CMB-sound-horizon-independent H_0 measurements partitioned into distance-ladder and one-step groups. The Kolmogorov–Smirnov test applied to the two populations returns $p = 0.0001$, meaning the probability they share a common parent distribution is negligible. The distance-ladder group yields $H_0 = 72.8 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ while the one-step group, after removing two known outliers, returns $H_0 \approx 68.3 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ —consistent with the sound-horizon-based CMB measurements. This shifts the framing from an early/late conflict to a ladder/non-ladder distinction, with implications for both systematic error searches and model building. Mörtzell et al. (2022) [2] place this conclusion on a quantitative footing for the Cepheid rung specifically: across all reasonable Cepheid colour-luminosity calibration variants, H_0 remains in 3.2σ – 4.7σ tension with *Planck*, and the total calibration-induced spread of $\approx 3.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ is a factor of ≈ 1.6 less than the gap to *Planck*. Riess et al. (2022) [3] deliver the highest-precision late-Universe measurement to date, $H_0 = 73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$, with comprehensive internal consistency checks showing that no single anchor, Cepheid subsample, or systematic effect is responsible for the discrepancy.

The confluence of these three papers supports a nuanced conclusion: the Hubble tension is unlikely to evaporate with improved statistics or better-characterised systematics within the existing framework. Moreover, the structural reframing offered by Perivolaropoulos (2024) [1] suggests that resolving the tension may require confronting why the distance ladder as a whole—across diverse calibrators and indicators—consistently returns elevated H_0 values relative to probes that bypass it entirely. The community’s attention should therefore focus on: (i) independent measurement pathways, particularly gravitational wave standard sirens and *JWST*-based distance indicators that bypass the Cepheid rung; (ii) theoretical models capable of globally satisfying all cosmological datasets including the one-step probes; and (iii) resolution of the SH0ES Cepheid vs. CCHP TRGB sub-controversy, which itself remains an important open question.

If the tension persists at its current significance with the next generation of observations, the implication will be that the most successful physical model of the observable Universe—flat Λ CDM—is incomplete in a fundamental way. Resolving it will require not merely better data, but a willingness to question assumptions that have underpinned cosmological modelling for three decades.

References

- [1] Perivolaropoulos, L. (2024). Hubble tension or distance ladder crisis? *Physical Review D*, **110**(12), 123518. [doi:10.1103/PhysRevD.110.123518](https://doi.org/10.1103/PhysRevD.110.123518) [arXiv:2408.11031]
- [2] Mörtzell, E., Goobar, A., Johansson, J., & Dhawan, S. (2022). Sensitivity of the Hubble constant determination to Cepheid calibration. *The Astrophysical Journal*, **935**, 58. [doi:10.3847/1538-4357/ac7c19](https://doi.org/10.3847/1538-4357/ac7c19)
- [3] Riess, A. G., Yuan, W., Macri, L. M., et al. (2022). A comprehensive measurement of the local value of the Hubble constant with $1 \text{ km s}^{-1} \text{ Mpc}^{-1}$ uncertainty from the Hubble Space Telescope and the SH0ES team. *The Astrophysical Journal Letters*, **934**, L7. [doi:10.3847/2041-8213/ac5c5b](https://doi.org/10.3847/2041-8213/ac5c5b)
- [4] Hubble, E. (1929). A relation between distance and radial velocity among extra-galactic nebulae. *Proceedings of the National Academy of Sciences*, **15**(3), 168–173. [doi:10.1073/pnas.15.3.168](https://doi.org/10.1073/pnas.15.3.168)
- [5] Trimble, V. (1996). H_0 : The incredible shrinking constant, 1925–1975. *Publications of the Astronomical Society of the Pacific*, **108**(721), 1073–1082. [doi:10.1086/133837](https://doi.org/10.1086/133837)
- [6] Freedman, W. L., Madore, B. F., Gibson, B. K., et al. (2001). Final results from the Hubble Space Telescope Key Project to measure the Hubble constant. *The Astrophysical Journal*, **553**(1), 47–72. [doi:10.1086/320638](https://doi.org/10.1086/320638)
- [7] Planck Collaboration; Aghanim, N., Akrami, Y., et al. (2020). Planck 2018 results. VI. Cosmological parameters. *Astronomy & Astrophysics*, **641**, A6. [doi:10.1051/0004-6361/201833910](https://doi.org/10.1051/0004-6361/201833910)
- [8] Alam, S., Aubert, M., Avila, S., et al. (2021). Completed SDSS-IV eBOSS: Cosmological implications. *Physical Review D*, **103**(8), 083533. [doi:10.1103/PhysRevD.103.083533](https://doi.org/10.1103/PhysRevD.103.083533)
- [9] Cooke, R. J., Pettini, M., & Steidel, C. C. (2018). One percent determination of the primordial deuterium abundance. *The Astrophysical Journal*, **855**(2), 102. [doi:10.3847/1538-4357/aaab53](https://doi.org/10.3847/1538-4357/aaab53)
- [10] Poulin, V., Smith, T. L., Karwal, T., & Kamionkowski, M. (2019). Early dark energy can resolve the Hubble tension. *Physical Review Letters*, **122**(22), 221301. [doi:10.1103/PhysRevLett.122.221301](https://doi.org/10.1103/PhysRevLett.122.221301)

- [11] Di Valentino, E., Melchiorri, A., & Silk, J. (2020). Reconciling Planck with the local value of H_0 in extended parameter space. *Physics Letters B*, **806**, 135555. [doi:10.1016/j.physletb.2020.135555](https://doi.org/10.1016/j.physletb.2020.135555)
- [12] Di Valentino, E., Mena, O., Pan, S., et al. (2021). In the realm of the Hubble tension—a review of solutions. *Classical and Quantum Gravity*, **38**(15), 153001. [doi:10.1088/1361-6382/ac086d](https://doi.org/10.1088/1361-6382/ac086d)
- [13] Efstathiou, G. (2021). To H_0 or not to H_0 ? *Monthly Notices of the Royal Astronomical Society*, **505**(3), 3866–3872. [doi:10.1093/mnras/stab1588](https://doi.org/10.1093/mnras/stab1588)
- [14] Leavitt, H. S., & Pickering, E. C. (1912). Periods of 25 variable stars in the Small Magellanic Cloud. *Harvard College Observatory Circular*, **173**, 1–3.
- [15] Pietrzyński, G., Graczyk, D., Gallette, A., et al. (2019). A distance to the Large Magellanic Cloud precise to one per cent. *Nature*, **567**(7747), 200–203. [doi:10.1038/s41586-019-0999-4](https://doi.org/10.1038/s41586-019-0999-4)
- [16] Reid, M. J., Pesce, D. W., & Riess, A. G. (2019). An improved distance to NGC 4258 and its implications for the Hubble constant. *The Astrophysical Journal Letters*, **886**(2), L27. [doi:10.3847/2041-8213/ab552d](https://doi.org/10.3847/2041-8213/ab552d)
- [17] Madore, B. F. (1982). The period-luminosity relation as a distance indicator. *The Astrophysical Journal*, **253**, 575–579. [doi:10.1086/159659](https://doi.org/10.1086/159659)
- [18] Guy, J., Astier, P., Baumont, S., et al. (2007). SALT2: Using distant supernovae to improve the use of Type Ia supernovae as distance indicators. *Astronomy & Astrophysics*, **466**(1), 11–21. [doi:10.1051/0004-6361:20066930](https://doi.org/10.1051/0004-6361:20066930)
- [19] Aiola, S., Calabrese, E., Maurin, L., et al. (2020). The Atacama Cosmology Telescope: DR4 maps and cosmological parameters. *Journal of Cosmology and Astroparticle Physics*, **2020**(12), 047. [doi:10.1088/1475-7516/2020/12/047](https://doi.org/10.1088/1475-7516/2020/12/047)
- [20] Birrer, S., Shajib, A. J., Galan, A., et al. (2020). TDCOSMO IV: Hierarchical time-delay cosmography. *Astronomy & Astrophysics*, **643**, A165. [doi:10.1051/0004-6361/202038861](https://doi.org/10.1051/0004-6361/202038861)
- [21] Freedman, W. L., Madore, B. F., Hoyt, T., et al. (2021). Measurements of the Hubble constant: Tensions in perspective. *The Astrophysical Journal*, **919**(1), 16. [doi:10.3847/1538-4357/ac0e95](https://doi.org/10.3847/1538-4357/ac0e95)
- [22] Blakeslee, J. P., Jensen, J. B., Ma, C.-P., Milne, P. A., & Greene, J. E. (2021). The Hubble constant from infrared surface brightness fluctuation distances. *The Astrophysical Journal*, **911**(1), 65. [doi:10.3847/1538-4357/abe86a](https://doi.org/10.3847/1538-4357/abe86a)

- [23] Riess, A. G., Macri, L. M., Hoffmann, S. L., et al. (2016). A 2.4% determination of the local value of the Hubble constant. *The Astrophysical Journal*, **826**(1), 56. [doi:10.3847/0004-637X/826/1/56](https://doi.org/10.3847/0004-637X/826/1/56)
- [24] Riess, A. G., Casertano, S., Yuan, W., et al. (2021). Cosmic distances calibrated to 1% precision with Gaia EDR3 parallaxes and HST photometry of 75 Milky Way Cepheids confirm tension with Λ CDM. *The Astrophysical Journal Letters*, **908**(1), L6. [doi:10.3847/2041-8213/abdbaf](https://doi.org/10.3847/2041-8213/abdbaf)
- [25] Fitzpatrick, E. L. (1999). Correcting for the effects of interstellar extinction. *Publications of the Astronomical Society of the Pacific*, **111**(755), 63–75. [doi:10.1086/316293](https://doi.org/10.1086/316293)
- [26] Gordon, K. D., Clayton, G. C., Misselt, K. A., Landolt, A. U., & Wolff, M. J. (2003). A quantitative comparison of SMC, LMC, and Milky Way extinction curves. *The Astrophysical Journal*, **594**(1), 279–293. [doi:10.1086/376774](https://doi.org/10.1086/376774)
- [27] Fausnaugh, M. M., Kochanek, C. S., Gerke, J. R., et al. (2015). Determining the Hubble constant using giant extragalactic H II regions. *Monthly Notices of the Royal Astronomical Society*, **450**(4), 3597–3614. [doi:10.1093/mnras/stv881](https://doi.org/10.1093/mnras/stv881)
- [28] Biswas, R., Goobar, A., Dhawan, S., et al. (2021). A new statistical method for cosmological analysis of Type Ia supernovae. *arXiv preprint*, arXiv:2103.16978. [arXiv:2103.16978](https://arxiv.org/abs/2103.16978)
- [29] Fitzpatrick, E. L., Massa, D., Gordon, K. D., Bohlin, R., & Clayton, G. C. (2019). An analysis of interstellar extinction curves. V. Infrared through ultraviolet. *The Astrophysical Journal*, **886**(2), 108. [doi:10.3847/1538-4357/ab4c3a](https://doi.org/10.3847/1538-4357/ab4c3a)
- [30] Tammann, G. A., Reindl, B., & Sandage, A. (2011). The Hubble constant and the expansion age of the Universe. *Astronomy & Astrophysics*, **531**, A134. [doi:10.1051/0004-6361/201016382](https://doi.org/10.1051/0004-6361/201016382)
- [31] Sullivan, M., Conley, A., Howell, D. A., et al. (2010). The dependence of Type Ia supernovae luminosities on their host galaxies. *Monthly Notices of the Royal Astronomical Society*, **406**(2), 782–802. [doi:10.1111/j.1365-2966.2010.16731.x](https://doi.org/10.1111/j.1365-2966.2010.16731.x)
- [32] Lindegren, L., Klioner, S. A., Hernández, J., et al. (2021). Gaia Early Data Release 3: The astrometric solution. *Astronomy & Astrophysics*, **649**, A2. [doi:10.1051/0004-6361/202039709](https://doi.org/10.1051/0004-6361/202039709)
- [33] Breuval, L., Riess, A. G., Kervella, P., Anderson, R. I., & Romaniello, M. (2022). Influence of the metallicity of Cepheids for the distance ladder calibration. *The Astrophysical Journal*, **939**(2), 89. [doi:10.3847/1538-4357/ac97e0](https://doi.org/10.3847/1538-4357/ac97e0)

- [34] Kenworthy, W. D., Scolnic, D., & Riess, A. (2019). The local perspective on the Hubble tension: Local structure does not impact measurement of H_0 . *The Astrophysical Journal*, **875**(2), 145. doi:10.3847/1538-4357/ab0ebf
- [35] Hill, J. C., McDonough, E., Toomey, M. W., & Alexander, S. (2020). Early dark energy does not restore cosmological concordance. *Physical Review D*, **102**(4), 043507. doi:10.1103/PhysRevD.102.043507
- [36] Scolnic, D., Brout, D., Carr, A., et al. (2022). The Pantheon+ analysis: The full data set and light-curve release. *The Astrophysical Journal*, **938**(2), 113. doi:10.3847/1538-4357/ac8b7a
- [37] Brout, D., Scolnic, D., Popovic, B., et al. (2022). The Pantheon+ analysis: Cosmological constraints. *The Astrophysical Journal*, **938**(2), 110. doi:10.3847/1538-4357/ac8e04
- [38] Soltis, J., Casertano, S., & Riess, A. G. (2021). The parallax of ω Centauri from Gaia EDR3 and a direct calibration of the TRGB and the Hubble constant. *The Astrophysical Journal Letters*, **908**(1), L5. doi:10.3847/2041-8213/abdbad
- [39] Yuan, W., Riess, A. G., Macri, L. M., Casertano, S., & Scolnic, D. M. (2019). Consistent calibration of the TRGB in the LMC on the HST photometric system and a redetermination of the Hubble constant. *The Astrophysical Journal*, **886**(1), 61. doi:10.3847/1538-4357/ab4bc9
- [40] Abbott, B. P., Abbott, R., Abbott, T. D., et al. (2017). A gravitational-wave standard siren measurement of the Hubble constant. *Nature*, **551**(7678), 85–88. doi:10.1038/nature24471
- [41] Chen, H.-Y., Fishbach, M., & Holz, D. E. (2018). A two per cent Hubble constant measurement from standard sirens within five years. *Nature*, **562**(7728), 545–547. doi:10.1038/s41586-018-0606-0